RESPONSIBLE AI

HW1: Algorithmic Fairness in Recidivism Prediction

Introduction

The COMPAS debate is often presented as a factual disagreement: either ProPublica was right that the tool is racially biased, or Northpointe was right that it is fair. My analysis reproduces the statistical patterns emphasized by each organization, but those patterns do not carry equal ethical weight. I side with ProPublica. When a prediction can affect bail, sentencing, or parole, the distribution of harmful errors should matter more than calibration within score categories. Northpointe's calibration defense is numerically accurate, but it does not address who bears the consequences when COMPAS is wrong.

What the Data Show

Three findings support my position. First, COMPAS has substantially different error rates for Black and white defendants. Among defendants who were not rearrested within two years, 42.3% of Black defendants were labeled Medium or High risk, compared with 22.0% of white defendants. Black defendants who were not rearrested were therefore almost twice as likely to receive an incorrect adverse label.

The pattern reverses among defendants who were rearrested. In this group, 49.6% of white defendants were incorrectly labeled Low risk, compared with 28.5% of Black defendants. Black defendants are more likely to experience the error that can result in stricter treatment, while white defendants are more likely to receive the less harmful error.

Second, the data support Northpointe's predictive-parity argument. Within the same decile score, Black and white defendants generally have similar rearrest rates. At decile 7, the rates are 60.9% for Black defendants and 60.2% for white defendants. The positive predictive value of a Medium-or-High label is also reasonably close: 65.0% for Black defendants and 59.5% for white defendants. Northpointe focused on whether a score had a similar meaning across racial groups, while ProPublica focused on whether prediction errors were distributed equally.

Third, the regression evidence strengthens ProPublica's position. After controlling for prior offenses, age, sex, and charge severity, Black defendants had 1.63 times the odds of receiving a Medium-or-High label compared with otherwise similar white defendants (95% CI: 1.42 to 1.86, $p < .001$). In a parallel model predicting two-year rearrest, the race effect was much smaller and not statistically significant (OR = 1.10, 95% CI: 0.98 to 1.25, $p = .119$).

The racial disparity in COMPAS labels remains larger than the disparity in rearrest across the controlled, reduced-control, and unadjusted models. This consistency suggests that the result is not driven solely by the selected controls, although it does not formally prove that the coefficients differ.

Why the Evidence Favors ProPublica

Northpointe's calibration argument is relevant, but it is not enough to establish fairness in this context. Calibration asks whether defendants with the same score have similar rearrest rates. It does not ask which groups are more likely to be harmed when the model makes a mistake. That distinction matters because a false-positive classification can contribute to detention or a harsher sentence. A model can be calibrated and still place that burden disproportionately on one racial group.

This conflict cannot simply be solved through better model tuning. The observed rearrest rates are 52.3% for Black defendants and 39.1% for white defendants. Chouldechova (2017) and Kleinberg et al. (2017) show that when base rates differ, calibration and equal error rates generally cannot both be achieved. A calibrated model will assign more high scores to the group with the higher observed outcome rate. Using the same threshold will then create different error rates. Reducing those differences usually weakens calibration. The system must therefore decide which form of fairness matters more.

Rawls's veil of ignorance supports ProPublica's position. Without knowing their race or which side of the false-positive gap they would occupy, people would be unlikely to accept a system in which one group has almost twice the chance of receiving an incorrect adverse label. Rawls's difference principle also questions an arrangement in which the most harmful error falls disproportionately on an already disadvantaged group.

Kant's respect-for-persons principle leads to a similar conclusion. Consistently assigning harsher labels to similarly situated members of one group risks treating those individuals as a means of preserving overall predictive performance. That tradeoff is difficult to justify when the cost is an individual's liberty.

The data also have an important limitation. The outcome measures rearrest, not proven reoffending. Rearrest may reflect differences in policing and surveillance as well as behavior. This weakens Northpointe's defense because calibration to a potentially distorted proxy does not prove fairness.

Operationalizing Fairness

The appropriate fairness standard should depend on the harm caused by an error. Calibration may be acceptable when mistakes are inexpensive or reversible. Bail and sentencing are different because a false positive can deprive someone of liberty.

Criminal-justice systems should therefore prioritize equal error rates, particularly false-positive parity.

My primary recommendation is that a COMPAS label should not independently justify detention, a longer sentence, or denial of parole unless an independent audit shows that the false-positive-rate gap between racial groups is below a publicly defined threshold. Courts or legislatures should establish that threshold before the tool is approved, disclose the justification for it, and reassess it through recurring audits. If COMPAS fails the standard, its score should not be used to increase punishment.

Jurisdictions should also require audits that report false-positive rates, false-negative rates, calibration, accuracy, and base rates for each group. Judges should document how a score affects their decision, and defendants should be allowed to examine and challenge the information used to calculate it.

Group-specific thresholds can reduce error-rate differences, but they may cause defendants with the same score to receive different recommendations based on race. A more defensible approach combines transparent audits, limits on how COMPAS may be used, meaningful human review, and rights to explanation and appeal.

Northpointe shows that COMPAS is approximately calibrated. ProPublica shows who bears the cost of maintaining that calibration. In a system capable of taking away liberty, the second question matters more. Fairness must be an explicit and accountable policy choice, not something assumed because a model is accurate on average.

Appendix: Supporting Tables

Classification rule. Medium or High is treated as a positive prediction; Low is treated as negative. The outcome is rearrest within two years, not proven reoffending.

Table A1. Observed Two-Year Rearrest Rate by Race

Race	N	Rearrest Rate
African-American	3,175	52.3%
Caucasian	2,103	39.1%

Table A2. Error Rates by Race

Race	False-Positive Rate	False-Negative Rate
African-American	42.3%	28.5%
Caucasian	22.0%	49.6%

Table A3. Predictive Parity by Race

Race	PPV for Medium/High Label	Rearrest Rate at Decile 7
African-American	65.0%	60.9% (209/343)
Caucasian	59.5%	60.2% (68/113)

Note. Similar PPV and decile-7 rearrest rates support Northpointe's predictive-parity claim.

Table A4. Logistic Regression with Full Controls

Outcome	OR for Black	95% CI	p-value
COMPAS Medium/High label	1.63	1.42 to 1.86	< .001
Observed two-year rearrest	1.10	0.98 to 1.25	.119

Note. Models control for priors_count, age_cat, sex, and c_charge_degree. Caucasian is the reference group. The nonsignificant rearrest result does not prove that the groups have identical rearrest risk. It means this analysis does not detect a difference after controlling for the available variables.

Table A5. Robustness Across Specifications

Specification	OR: COMPAS Label	OR: Rearrest	OR Ratio
Full controls	1.63 ($p < .001$)	1.10 ($p = .119$)	1.47
Without priors_count	2.30 ($p < .001$)	1.47 ($p < .001$)	1.57
Race only	2.75 ($p < .001$)	1.71 ($p < .001$)	1.61

Note. The race odds ratio is consistently larger in the COMPAS-label model than in the observed-rearrest model. The ratio remains approximately 1.5 to 1.6 across specifications, suggesting that the pattern is not produced solely by the selected control variables. The ratio is descriptive and is not a formal statistical test of coefficient equality.

Definitions and Data

FPR: Share of defendants not rearrested who were labeled Medium/High.

FNR: Share of defendants rearrested who were labeled Low.

PPV: Share of defendants labeled Medium/High who were rearrested.

Sample and Source: n = 5,278, including 3,175 African-American and 2,103 Caucasian defendants. Data: compas-scores.csv, a cleaned subsample of ProPublica's COMPAS dataset.

References

Chouldechova, A. (2017). Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data, 5*(2), 153–163.

Kleinberg, J., Mullainathan, S., & Raghavan, M. (2017). Inherent trade-offs in the fair determination of risk scores. *Proceedings of Innovations in Theoretical Computer Science*.